

Duality Theory Notes

Math Camp 2026 — Teaching Notes

Primal problem: choose $\mathbf{x}$ to minimize cost subject to constraints.

Dual problem: find the best lower bound on that minimum using shadow prices on constraints.

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1 Why duality?

Many economic problems are **minimization** problems: choose activity levels $\mathbf{x}$ to minimize cost $\mathbf{c}^\top \mathbf{x}$ subject to requirements $A\mathbf{x} \geq \mathbf{b}$.

Key idea. Before solving exactly, ask: *what is the best lower bound we can prove on minimum cost?* Duality turns that bound into a **second optimization problem** whose variables $\mathbf{y}$ are **shadow prices** on the constraints.

Economics. The **primal** is a planner's cost-minimization problem; the **dual** is a price-setter's valuation problem. Any feasible prices $\mathbf{y} \geq 0$ with $A^\top \mathbf{y} \leq \mathbf{c}$ give a valid resource value $\mathbf{b}^\top \mathbf{y}$ that no feasible plan can beat. Duality asks for the **tightest** such bound. This underlies supporting prices, zero-profit conditions, and competitive price interpretations in graduate micro.

2 Linear programming duality

This section builds LP duality from one symmetric primal–dual pair, then adds the main theorems.

2.1 The primal–dual pair

The **primal** (resource-requirement form):

$$(P) \quad \min \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}. \quad (1)$$

Derivation of the dual. Pick any $\mathbf{y} \geq \mathbf{0}$ and add the constraints: $\mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$. Require $\mathbf{c} \geq A^\top \mathbf{y}$ so that, for all $\mathbf{x} \geq \mathbf{0}$, $\mathbf{c}^\top \mathbf{x} \geq \mathbf{y}^\top A\mathbf{x}$. With primal feasibility, $\mathbf{c}^\top \mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$, so $\mathbf{b}^\top \mathbf{y}$ is a **lower bound** on the optimum. Maximise it to get the **dual**:

$$(D) \quad \max \mathbf{b}^\top \mathbf{y} \quad \text{s.t.} \quad A^\top \mathbf{y} \leq \mathbf{c}, \mathbf{y} \geq \mathbf{0}. \quad (2)$$

Economics. Read the pair as follows.

- x_j : scale of activity j ; c_j : unit cost.
- $(A\mathbf{x})_i \geq b_i$: requirement i (labor, output, delivery).
- y_i : **shadow price** of constraint i .
- $A^\top \mathbf{y} \leq \mathbf{c}$: imputed resource cost of each activity $\leq$ its direct cost.
- $\mathbf{b}^\top \mathbf{y}$: total value of requirements at prices $\mathbf{y}$.

Mini example. $\min 3x_1 + 5x_2$ s.t. $x_1 + 2x_2 \geq 4, x_1, x_2 \geq 0$ (meet 4 service units at minimum cost). Dual: $\max 4y$ s.t. $y \leq 3, 2y \leq 5, y \geq 0 \Rightarrow y^* = 3, \text{ cost} = 12$.

2.2 Weak duality

Theorem 1 (Weak duality). *For every feasible $\mathbf{x}$ of (1) and $\mathbf{y}$ of (2), $\mathbf{c}^\top \mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$, hence $p^* \geq d^*$.*

Proof. $A^\top \mathbf{y} \leq \mathbf{c}$ and $\mathbf{x} \geq \mathbf{0} \Rightarrow \mathbf{c}^\top \mathbf{x} \geq \mathbf{y}^\top A\mathbf{x}$; $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{y} \geq \mathbf{0} \Rightarrow \mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$. □

Remark 2 (Optimality certificate). *If feasible $\mathbf{x}, \mathbf{y}$ satisfy $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$, both are optimal.*

Economics. Cost is always at least resource value. Equality is a **certificate of efficiency**: the plan and prices are mutually consistent.

2.3 Farkas' Lemma

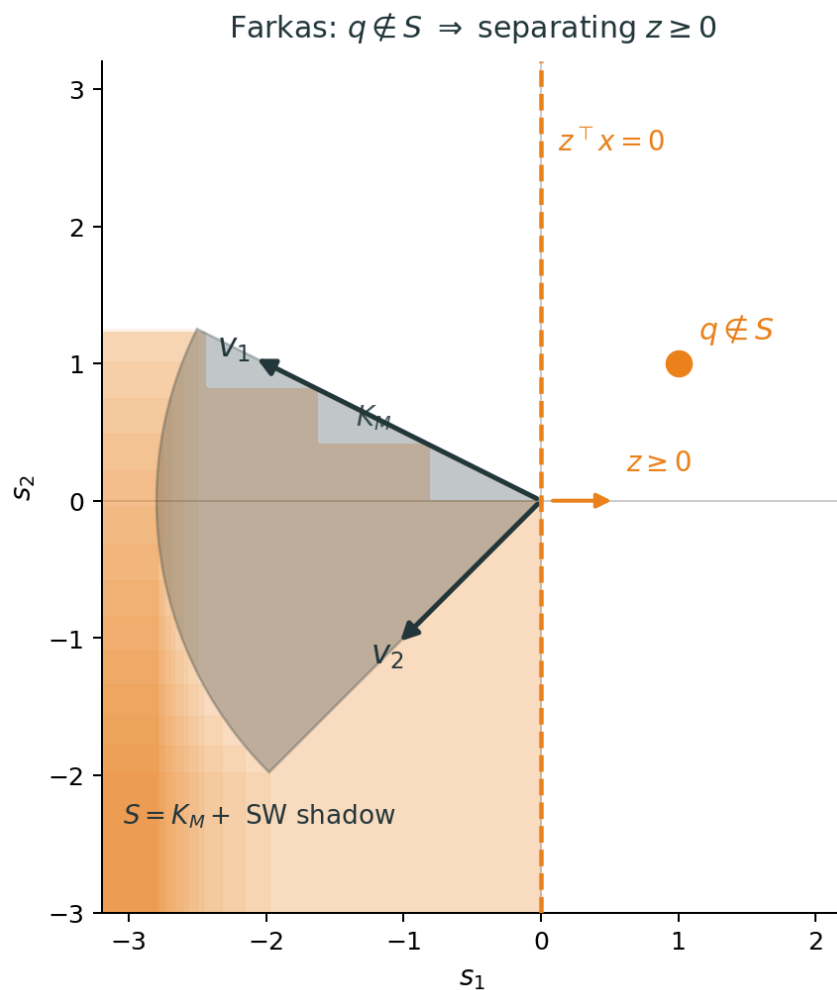
Strong duality uses the following theorem of alternatives.

Theorem 3 (Farkas' Lemma — inequality form). *Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{q} \in \mathbb{R}^m$. Exactly one of the following has a solution:*

- (i) $\exists \mathbf{x} \geq \mathbf{0}$ with $M\mathbf{x} \geq \mathbf{q}$;
- (ii) $\exists \mathbf{z} \geq \mathbf{0}$ with $M^T \mathbf{z} \leq \mathbf{0}$ and $\mathbf{q}^T \mathbf{z} > 0$.

Geometry. Let $S = \{M\mathbf{x} - \mathbf{y} : \mathbf{x}, \mathbf{y} \geq \mathbf{0}\} = K_M + (-\mathbb{R}_+^m)$, the column cone K_M plus free disposal in every negative coordinate direction.

- (i) holds $\Leftrightarrow \mathbf{q} \in S$.
- (ii) holds $\Leftrightarrow \mathbf{q} \notin S$; a separating normal $\mathbf{z} \geq \mathbf{0}$ certifies this.



Proof sketch. If both hold: $0 < \mathbf{q}^T \mathbf{z} \leq \mathbf{z}^T M\mathbf{x} \leq 0$. If (i) fails: $\mathbf{q} \notin S$; separation gives $\mathbf{z}$ with $\mathbf{z}^T \mathbf{s} \leq 0$ for all $\mathbf{s} \in S$ and $\mathbf{z}^T \mathbf{q} > 0$; checking $-\mathbf{e}_i$, $M\mathbf{x} \in S$ gives $\mathbf{z} \geq \mathbf{0}$ and $M^T \mathbf{z} \leq \mathbf{0}$.

Economics. Either a target is feasible with free disposal, or a nonnegative price vector exposes it as too ambitious—a no-arbitrage statement.

2.4 Strong duality

Theorem 4 (Strong duality for LP). *If (1) has an optimal value p^* , then (2) has $d^* = p^*$.*

Proof. For any $\varepsilon > 0$, no $\mathbf{x} \geq \mathbf{0}$ satisfies $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{c}^\top \mathbf{x} \leq p^* - \varepsilon$. Write $M\mathbf{x} \geq \mathbf{q}$ with $M = \begin{pmatrix} A \\ -\mathbf{c}^\top \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} \mathbf{b} \\ -(p^* - \varepsilon) \end{pmatrix}$. By Farkas, $\exists \mathbf{y} \geq \mathbf{0}, \mu \geq 0$ with $A^\top \mathbf{y} - \mu \mathbf{c} \leq \mathbf{0}$ and $\mathbf{b}^\top \mathbf{y} - \mu(p^* - \varepsilon) > 0$. If $\mu = 0$, primal feasibility contradicts $\mathbf{b}^\top \mathbf{y} > 0$; so $\mu > 0$. With $\tilde{\mathbf{y}} = \mathbf{y}/\mu$: dual feasible and $\mathbf{b}^\top \tilde{\mathbf{y}} > p^* - \varepsilon$. Let $\varepsilon \downarrow 0$; weak duality gives $d^* = p^*$. $\square$

Economics. Minimum cost = maximum resource value at optimality. Efficient allocations can be supported by shadow prices.

2.5 Complementary slackness

Theorem 5 (Complementary slackness). *Feasible $\mathbf{x}, \mathbf{y}$ are optimal iff*

$$(A\mathbf{x} - \mathbf{b})_i y_i = 0 \quad \forall i, \quad (c - A^\top \mathbf{y})_j x_j = 0 \quad \forall j.$$

In words.

If ...	then ...
primal constraint i is slack	$y_i = 0$
$y_i > 0$	primal constraint i is tight
dual constraint j is slack	$x_j = 0$
$x_j > 0$	dual constraint j is tight

Necessity. Optimality $\Rightarrow \mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$ (strong duality). Weak duality forces both inequalities in the chain to be equalities, hence both dot products vanish; nonnegativity then forces each product term to be zero. $\square$

Sufficiency. Complementary slackness gives $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$; weak duality certifies optimality. $\square$

Remark 6 (Direct proof). *From $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$ and weak duality: any feasible $\tilde{\mathbf{x}}$ has $\mathbf{c}^\top \tilde{\mathbf{x}} \geq \mathbf{c}^\top \mathbf{x}$; any feasible $\tilde{\mathbf{y}}$ has $\mathbf{b}^\top \tilde{\mathbf{y}} \leq \mathbf{b}^\top \mathbf{y}$.*

Economics. Zero-profit / full-utilization logic. Unused capacity $\Rightarrow$ zero shadow price; scarce resource $\Rightarrow$ binds. Unprofitable activity $\Rightarrow x_j = 0$; used activity $\Rightarrow$ breaks even at shadow prices. Links to the Day 3 producer-theory slide (efficient production supported by prices when free disposal holds).

2.6 Dual of a general LP

Key idea. Same bounding argument; signs follow from constraint and variable types.

Claim. For a minimisation with any mix of $\geq, \leq, =$ constraints and nonnegative, nonpositive, or free variables: dual is a **maximisation** in y_i (one per primal constraint). If the primal is a **maximisation**, reverse every inequality below.

Proof. $\mathbf{c}^\top \mathbf{x} - \mathbf{b}^\top \mathbf{y} = (\mathbf{c} - A^\top \mathbf{y})^\top \mathbf{x} + \mathbf{y}^\top (A\mathbf{x} - \mathbf{b})$. Require each part ≥ 0 for all feasible $\mathbf{x}$; maximise $\mathbf{b}^\top \mathbf{y}$.

Rules (primal min $\Rightarrow$ dual max).

Primal	Dual
$\mathbf{a}_i^\top \mathbf{x} \geq b_i$	$y_i \geq 0$
$\mathbf{a}_i^\top \mathbf{x} \leq b_i$	$y_i \leq 0$
$\mathbf{a}_i^\top \mathbf{x} = b_i$	y_i free
$x_j \geq 0$	$(A^\top \mathbf{y})_j \leq c_j$
$x_j \leq 0$	$(A^\top \mathbf{y})_j \geq c_j$
x_j free	$(A^\top \mathbf{y})_j = c_j$

Example.

$$\min 3x_1 - 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 5, \quad 2x_1 - x_2 \leq 10, \quad x_1 - 3x_2 = 4, \quad x_1 \geq 0, \quad x_2 \text{ free.}$$

Dual:

$$\max 5y_1 + 10y_2 + 4y_3 \quad \text{s.t.} \quad y_1 + 2y_2 + y_3 \leq 3, \quad y_1 - y_2 - 3y_3 = -2, \quad y_1 \geq 0, \quad y_2 \leq 0, \quad y_3 \text{ free.}$$

3 Lagrangian duality

LP duality is the special case of a general construction: multiply constraints by weights, add to the objective, and maximise the resulting lower bound.

3.1 Setup

$$(P) \quad \min f_0(\mathbf{x}) \quad \text{s.t.} \quad f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p, \quad \mathbf{x} \in D.$$

Optimal value: p^* .

3.2 Lagrangian and dual problem

Lagrangian.

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}), \quad \boldsymbol{\lambda} \geq \mathbf{0}, \quad \boldsymbol{\nu} \text{ free.}$$

For feasible $\mathbf{x}$: $L \leq f_0$ (added terms are ≤ 0).

Dual function. $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in D} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$.

Dual problem.

$$(D) \quad \max g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \quad \text{s.t.} \quad \boldsymbol{\lambda} \geq \mathbf{0}.$$

Always $d^* \leq p^*$ (**weak duality**); $p^* - d^*$ is the **duality gap**.

3.3 Geometry I: one inequality — the (u, t) cloud

Key idea. Dual = best lower bound from a supporting **line** below the whole achievable set $\mathcal{A} \subset \mathbb{R}^2$.

Problem.

$$\min_{\mathbf{x}} f_0(\mathbf{x}) \quad \text{s.t.} \quad f_1(\mathbf{x}) \leq 0.$$

Let $u = f_1(\mathbf{x})$, $t = f_0(\mathbf{x})$. As $\mathbf{x}$ runs over D ,

$$\mathcal{A} = \{ (u, t) : u = f_1(\mathbf{x}), t = f_0(\mathbf{x}), \mathbf{x} \in D \}.$$

$\mathcal{A}$ is the **whole cloud** of achievable pairs—feasible ($u \leq 0$) and infeasible.

Primal. Among $(u, t) \in \mathcal{A}$ with $u \leq 0$, minimise t . That value is p^* .

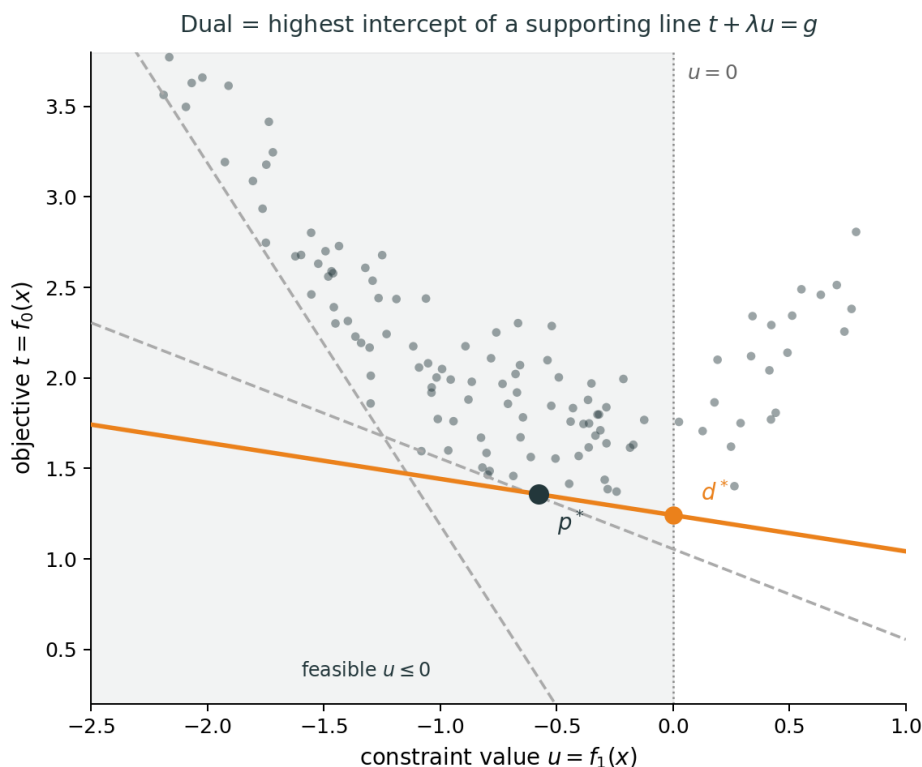
Dual function. For $\lambda \geq 0$, lines $t + \lambda u = \alpha$ have slope $-\lambda$ in the (u, t) plane.

$$g(\lambda) = \inf_{(u,t) \in \mathcal{A}} (t + \lambda u)$$

is the smallest α such that the line still touches $\mathcal{A}$ from below—its intercept on $u = 0$. The infimum is over **all** of $\mathcal{A}$, not only the feasible slice $u \leq 0$.

Why $g(\lambda) \leq p^*$. For any feasible point ($u \leq 0$): $t + \lambda u \leq t$ since $\lambda \geq 0$. Also $g(\lambda) \leq t + \lambda u$. Hence $g(\lambda) \leq t$, so $g(\lambda) \leq p^*$.

Dual problem. $d^* = \sup_{\lambda \geq 0} g(\lambda)$: tilt the supporting line (choose λ) to maximise the intercept at $u = 0$.



3.4 Geometry II: one inequality and one equality — the (u, v, t) cloud

Key idea. Adding an equality lets the supporting object tilt in a **third** direction; free ν can tighten the bound.

Problem.

$$\min_{\mathbf{x}} f_0(\mathbf{x}) \quad \text{s.t.} \quad f_1(\mathbf{x}) \leq 0, \quad h(\mathbf{x}) = 0.$$

Let $u = f_1(\mathbf{x})$, $v = h(\mathbf{x})$, $t = f_0(\mathbf{x})$.

$$\mathcal{A} = \{(u, v, t) : u = f_1(\mathbf{x}), v = h(\mathbf{x}), t = f_0(\mathbf{x}), \mathbf{x} \in D\}.$$

Again, $\mathcal{A}$ contains **all** achievable triples. Feasible points have $u \leq 0$, $v = 0$.

Primal. Among $(u, v, t) \in \mathcal{A}$ with $u \leq 0$ and $v = 0$, find the smallest $t \Rightarrow p^*$.

Dual function. Choose $\lambda \geq 0$ (for u) and ν free (for v).

$$g(\lambda, \nu) = \inf_{(u, v, t) \in \mathcal{A}} (t + \lambda u + \nu v).$$

The plane $t + \lambda u + \nu v = \alpha$ supports $\mathcal{A}$ from below; $g(\lambda, \nu)$ is its intercept at $u = 0$, $v = 0$.

Why $g(\lambda, \nu) \leq p^*$. For feasible (u, v, t) ($u \leq 0$, $v = 0$): $t + \lambda u + \nu v = t + \lambda u \leq t$. Since $g(\lambda, \nu) \leq t + \lambda u + \nu v$, we get $g(\lambda, \nu) \leq p^*$.

Dual problem. $d^* = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$: the best intercept from a hyperplane below the full cloud $\mathcal{A}$.

Why $\nu = 0$ can be weaker. With $\nu = 0$, the plane $t + \lambda u = \alpha$ cannot tilt in the v -direction. The infimum $g(\lambda, 0)$ may occur at $v \neq 0$, giving a low intercept at $v = 0$. Allowing ν to vary lets the plane tilt so the minimum over $\mathcal{A}$ can occur nearer $v = 0$, often raising the bound; for convex problems this can achieve $d^* = p^*$.

For LPs, $\mathcal{A}$ is a polyhedron (convex) $\Rightarrow$ strong duality when the primal optimum is finite.

3.5 LP recovers Section 2

Write $\min \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$ as $f_1 = \mathbf{b} - A\mathbf{x} \leq \mathbf{0}$, $f_2 = -\mathbf{x} \leq \mathbf{0}$:

$$L = \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{b} - A\mathbf{x}) + \boldsymbol{\mu}^\top (-\mathbf{x}).$$

$g(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} [(\mathbf{c} - A^\top \boldsymbol{\lambda} - \boldsymbol{\mu})^\top \mathbf{x} + \boldsymbol{\lambda}^\top \mathbf{b}]$. Finite g requires $A^\top \boldsymbol{\lambda} \leq \mathbf{c}$; then $g = \boldsymbol{\lambda}^\top \mathbf{b}$. With $\mathbf{y} = \boldsymbol{\lambda}$: the LP dual (2).

3.6 Strong duality for convex problems

Problem is **convex** if f_0, f_i are convex, h_j are affine, D is convex.

Theorem 7 (Strong duality). *If the primal is convex, **Slater's condition** holds (strict feasibility: $f_i(\mathbf{x}) < 0$, $h_j(\mathbf{x}) = 0$ for some $\mathbf{x} \in \text{relint } D$), and p^* is finite, then $d^* = p^*$ and the dual optimum is attained.*

Remark 8 (What is $\text{relint } D$?). The **relative interior** $\text{relint } D$ is the “interior” of D measured inside the flat (affine hull) that contains D , not inside all of $\mathbb{R}^n$.

- If $D = \mathbb{R}^n$, then $\text{relint } D = \mathbb{R}^n$.
- If D is a line segment in $\mathbb{R}^2$, its usual topological interior is empty, but $\text{relint } D$ is the **open segment** (endpoints excluded).
- If $D = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$ is a full-dimensional polyhedron, $\text{relint } D$ is the set of points that satisfy every inequality **strictly** ($A\mathbf{x} < \mathbf{b}$), i.e. not on a lower-dimensional face.

Slater asks for a **strictly feasible** point that is not trapped on a boundary of D : inequalities hold with slack ($f_i(\mathbf{x}) < 0$), equalities hold exactly, and $\mathbf{x}$ lies in the relative interior of the domain.

For LPs, a feasible point with finite optimum suffices. Slater is the convex analogue of “interior point exists.”

Economics. λ_i, ν_j are shadow prices on constraints. The dual finds the price system giving the best accounting lower bound on optimal cost. When the gap vanishes, prices fully explain the optimum.